# ELEC5206 Laboratory 2 Report

Characterization of a Photovoltaic Module

Bench: 4 | Attendance: Wednesday morning, 19 August 2026
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## 1. Aim and experimental method

The experiment characterized a photovoltaic (PV) module using a selectable resistive load and a buck converter with a fixed load. The objectives were to obtain I-V and P-V curves, estimate open-circuit voltage and short-circuit current, and identify the highest recorded power point in each sweep. The converter test also examined how PWM duty cycle changes the effective load seen by the PV module.

### 1.1 Equipment and measurement arrangement

The setup comprised a PV panel under halogen lamps, a 10 W resistor panel (2-220 Ω), a pre-engineered buck converter, Rigol function generator, 12 V DC driver supply, Tektronix MDO34 oscilloscope, and Lab-Volt DACI 9063 with LVDAC-EMS. E1 measured PV-terminal voltage, I1 measured series PV current, and PQS1 recorded power. The oscilloscope monitored the PWM input and converter output.

### 1.2 Procedure and operating settings

1. With lamps and auxiliary supply off, connect the PV module, current channel and selected load in series; connect the voltage channel across the PV terminals. Keep the lamp position fixed. The prescribed safety sequence requires tutor approval before energizing and avoiding hot lamps, panels and resistors.
2. For the physical-load test, illuminate the panel and allow one minute before acquisition. Sweep the selectable loads from 2 Ω to 220 Ω, recording resistance, voltage, current and power at each settled point; also obtain short- and open-circuit readings. The saved table contains 12 points, ordered by increasing voltage (Table 1).
3. For the converter test, connect PV positive to Vin (P1), the 2 Ω load to Vo (P2), and PV negative/load return to Gnd (P3). Supply the driver through J4 at 12 V with correct polarity. Check a nominal 0-5 V, 100 kHz PWM signal before connecting it to J3 pin 5, with common ground; J3 pin 4 is unused.
4. After the prescribed checks and one-minute illumination delay, enable the driver supply. Set duty to 10%, 15%, 20%, then 22-40% in 2-percentage-point steps and 50-90% in 10-point steps. Record the 18 voltage/current/power triples (Table 2), then plot both characteristic curves and identify the sampled peak.

### 1.3 Data treatment and theoretical reference

Recorded voltage, current and power are listed in Tables 1-2. The calculated product $P_{\mathrm{calc}}$ is included for comparison. For steady DC and an ideal lossless buck converter in continuous conduction:

$$P_{\mathrm{calc}} = V_{\mathrm{pv}} I_{\mathrm{pv}} ; \quad V_{\mathrm{o}} = D V_{\mathrm{pv}} ; \quad R_{\mathrm{in,ideal}} = \frac{R_{\mathrm{L}}}{D^2}$$

Here D is the duty-cycle fraction (26% = 0.26), and $R_{\mathrm{L}}$ = 2 Ω.

## 2. Results

Tables 1-2 contain the original measurements and the calculated product $P_{calc}$ = VI. Figures 1-2 show the corresponding LVDAC-EMS graphs. The original I-V graphs display only zero on the current axis; numerical current values are given in the tables.

*Table 1. Physical-resistor measurements.*

| R setting (Ω) | V (V) | I (A) | Recorded P (W) | $P_{calc}$ (W) |
|---|---|---|---|---|
| 0 | 0.144 | 0.401 | 0.014 | 0.058 |
| 2 | 0.939 | 0.407 | 0.379 | 0.382 |
| 10 | 3.989 | 0.396 | 1.579 | 1.580 |
| 22 | 8.269 | 0.38 | 3.142 | 3.142 |
| 27 | 9.843 | 0.367 | 3.614 | 3.612 |
| 30 | 10.97 | 0.36 | 3.952 | 3.949 |
| 33 | 11.65 | 0.353 | 4.116 | 4.112 |
| 40 | 13.81 | 0.343 | 4.728 | 4.737 |
| 82 | 17.45 | 0.212 | 3.7 | 3.699 |
| 150 | 17.83 | 0.117 | 2.086 | 2.086 |
| 220 | 17.92 | 0.08 | 1.425 | 1.434 |
| Not recorded | 18.23 | 0.007 | -0.047 | 0.128 |

*Table 2. Buck-converter measurements with a 2 Ω load.*

| Duty setting (%) | V (V) | I (A) | Recorded P (W) | $P_{calc}$ (W) |
|---|---|---|---|---|
| 10 | 17.7 | 0.058 | 1.007 | 1.027 |
| 15 | 16.83 | 0.13 | 2.185 | 2.188 |
| 20 | 16.05 | 0.229 | 3.669 | 3.675 |
| 22 | 15.57 | 0.271 | 4.217 | 4.219 |
| 24 | 15.12 | 0.314 | 4.747 | 4.748 |
| 26 | 14.04 | 0.344 | 4.826 | 4.830 |
| 28 | 12.43 | 0.35 | 4.345 | 4.350 |
| 30 | 11.06 | 0.356 | 3.936 | 3.937 |
| 32 | 9.919 | 0.362 | 3.591 | 3.591 |
| 34 | 9.001 | 0.368 | 3.313 | 3.312 |
| 36 | 8.161 | 0.374 | 3.053 | 3.052 |
| 38 | 7.472 | 0.381 | 2.844 | 2.847 |
| 40 | 6.954 | 0.393 | 2.729 | 2.733 |
| 50 | 4.661 | 0.405 | 1.885 | 1.888 |
| 60 | 3.27 | 0.406 | 1.327 | 1.328 |
| 70 | 2.404 | 0.409 | 0.981 | 0.983 |
| 80 | 1.831 | 0.412 | 0.752 | 0.754 |
| 90 | 1.415 | 0.412 | 0.58 | 0.583 |

## 2.1 Characteristic curves and sampled maxima

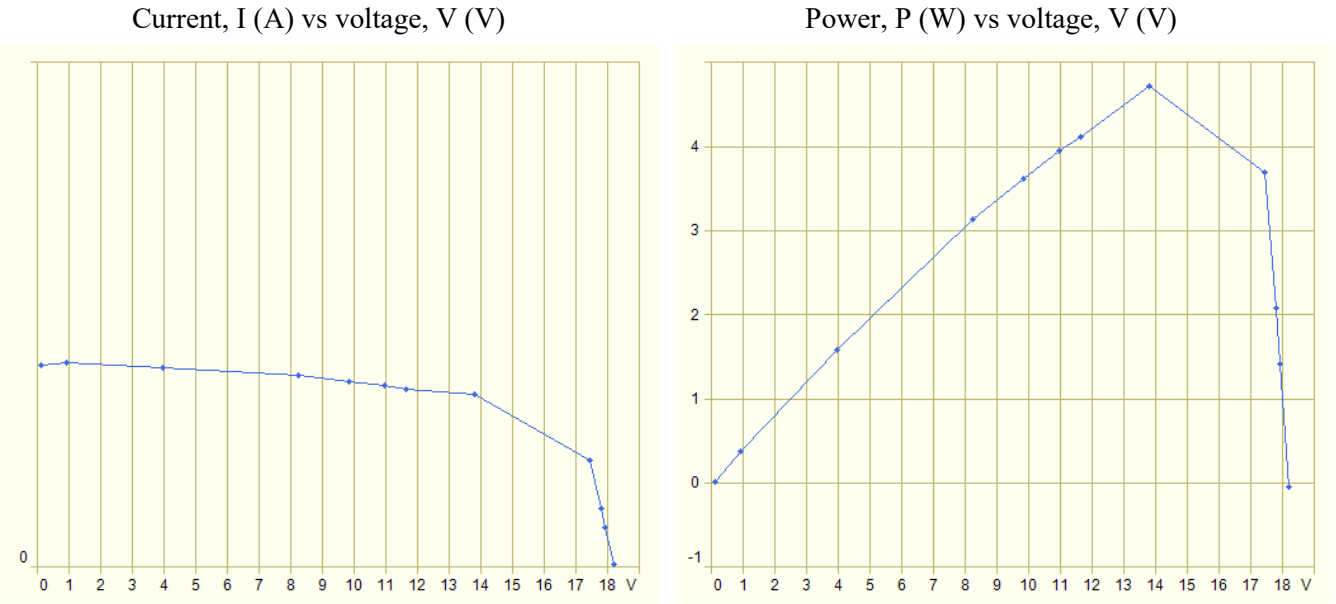



*Figure 1. Physical-resistor I-V and P-V curves from LVDAC-EMS.*

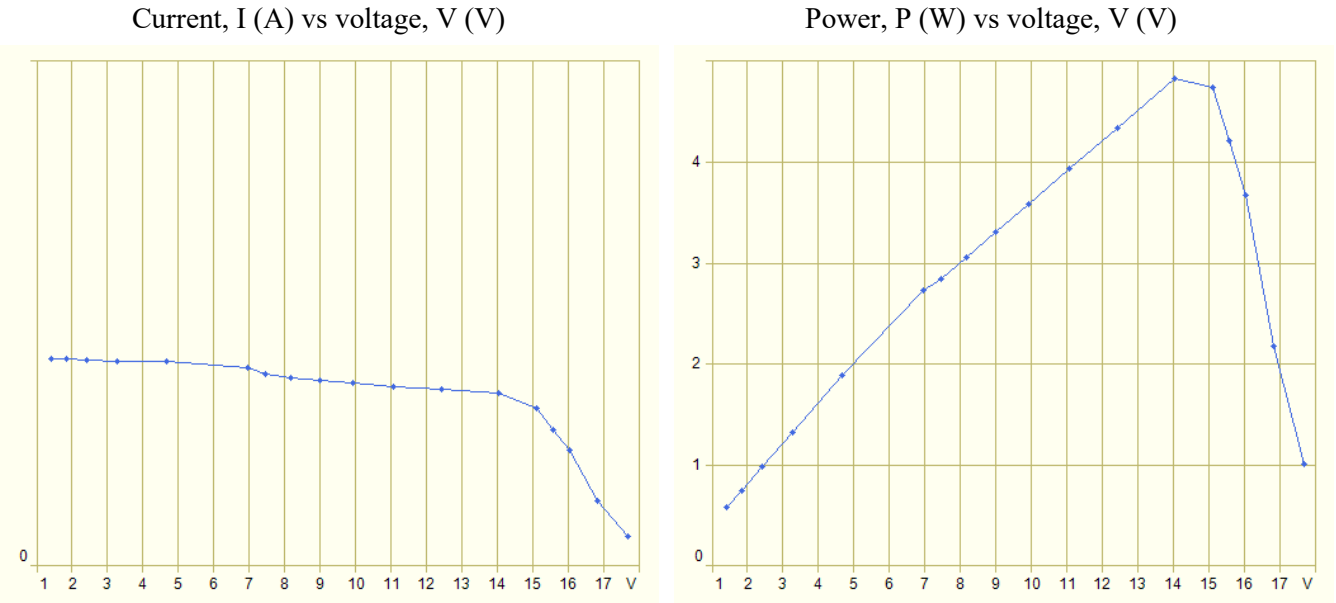



*Figure 2. Buck-converter I-V and P-V curves from LVDAC-EMS.*

*Table 3. Maximum recorded power for each method.*

| Method | Setting | V at peak (V) | I at peak (A) | Peak P (W) |
|---|---|---|---|---|
| Physical resistors | 40 Ω | 13.81 | 0.343 | 4.728 |
| Buck converter | 26% | 14.04 | 0.344 | 4.826 |

The resistor sweep gives $V_{oc} \approx 18.23$ V and $I_{sc} \approx 0.401$ A. The final load setting was not recorded; this point is treated as near-open circuit because its current was only 0.007 A. The nominal short-circuit voltage was 0.144 V, so both endpoint estimates are approximate.

Power in the buck test increased from 1.007 W at 10% duty to 4.826 W at 26%, then decreased to 0.580 W at 90%. The peak is the highest measured point in the sweep; a finer duty step could locate it more closely.

# 3. Discussion: measurement validity and load control

## 3.1 Record consistency and instrument limitations

The saved E1/I1 headers show AC, although the handout specifies DC. LVDAC distinguishes mean and RMS readings, so the acquisition mode needs checking before treating these values as DC means. With ripple, average power can also differ from the product of RMS voltage and current. The duty percentages in Table 2 are generator settings recorded during the test, rather than verified scope measurements.

For the ten finite non-zero resistor loads, the largest difference between recorded power and VI was 0.00883 W. The nominal-short and near-open differences were −0.04374 W and −0.17461 W, too large to attribute to display rounding alone. The near-open value of −0.047 W is therefore an anomalous reading. In the buck test, the largest difference was 0.0196 W; at the peak, VI = 4.830 W, close to the recorded 4.826 W.

## 3.2 Comparison of physical and virtual resistance

The converter peak is 0.098 W (2.07%) above the physical-load peak. This is not a converter-efficiency measurement or proof of increased available PV power: the tests were separate, with no irradiance/temperature trace or repeatability estimate. Manual resistance changes are simple but discrete and slow; PWM offers finer electronic adjustment and could be automated. Both methods place their sampled peak near 14 V and 0.344 A.

$$R_{\text{in,recorded}} = \frac{14.04\ \text{V}}{0.344\ \text{A}} = 40.81\ \Omega\,; \quad R_{\text{in,ideal}} = \frac{2\ \Omega}{0.26^2} = 29.59\ \Omega$$

The ideal model explains the direction of control: increasing D reduces input resistance, lowering PV voltage while increasing current. It does not quantitatively reproduce the recorded ratio at the peak. Unverified DC mode, converter non-idealities, actual gate timing, component values and conduction mode remain possible contributors; none is isolated by these records.

## 3.3 Oscilloscope evidence

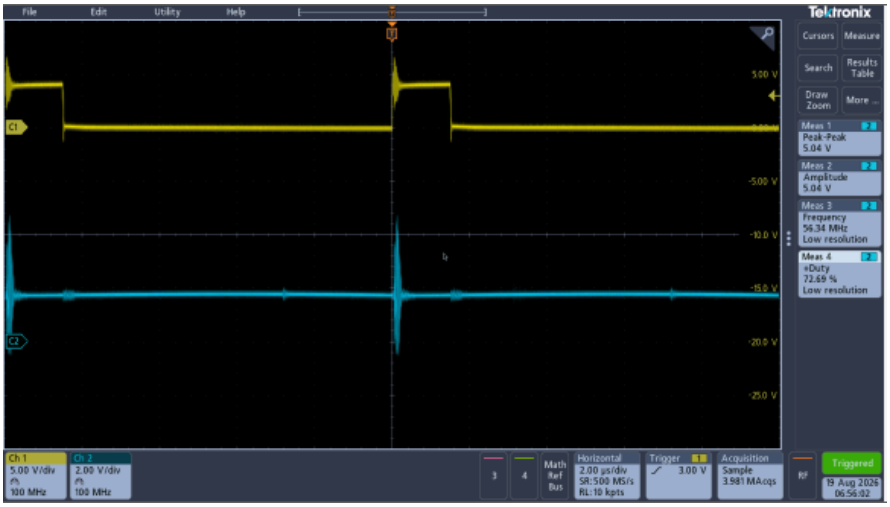



*Figure 3. Oscilloscope display: CH1 (yellow) and CH2 (cyan).*

At 2.00 µs/div, CH1 pulses are about five divisions apart, giving T ≈ 10 µs and f ≈ 100 kHz. The automatic 56.34 MHz and 27.69% readings belong to CH2 and show “Low resolution”, so they cannot verify PWM frequency or duty. Reliable output mean and ripple values were not obtained from these captures.

# 4. Thermal effects, MPPT and improvements

## 4.1 Voltage drift and the one-minute delay

Continued halogen exposure heats the module. For a typical silicon PV panel, increasing cell temperature lowers open-circuit and maximum-power voltage under comparable illumination; thus later tests can show lower voltage. Temperature mainly shifts the voltage operating point. This is an explanation of the expected mechanism, not a measured temperature trend in this dataset. The one-minute delay lets the lamp output and initial electrical transient settle, but does not guarantee thermal equilibrium.

Future tests should keep lamp distance/angle fixed, measure module temperature and irradiance with timestamps, and automate a short sweep after electrical settling. Repeat sweeps from a consistent initial temperature, use controlled cooling if available, and compare forward/reverse sweeps to expose drift. A repeatable starting condition is preferable to an arbitrary longer warm-up.

## 4.2 Buck converter with microcontroller MPPT

A microcontroller can sample calibrated PV voltage and current, calculate power, and update a bounded PWM duty command. Start conservatively, allow settling, then apply a small duty perturbation. If power rises, continue in the same direction; if it falls, reverse direction. Average synchronized samples to suppress switching noise and reduce step size near the peak. This perturb-and-observe method tracks changing conditions, but oscillates around the optimum and can be confused by rapidly changing irradiance.

The data suggest 26% as an initial search region, not a permanent optimum. Increasing buck duty moves the PV operating voltage down in this setup. Use current/voltage limits, undervoltage protection and driver-compatible PWM, rather than connecting a microcontroller pin directly to a MOSFET gate. An incremental-conductance alternative seeks the steady-state MPP condition:

$$\frac{\mathrm{d}P}{\mathrm{d}V} = I + V\frac{\mathrm{d}I}{\mathrm{d}V} = 0$$

## 4.3 Learning and recommended improvements

The exercise links load-line movement, PV characteristics and converter control: maximum current or voltage alone does not maximize power. Priorities for a repeat are verified DC acquisition, zero/offset checks, corrected scope measurement sources, finer sampling around the peak, repeated synchronized measurements, and logging output voltage/current to assess conversion efficiency. Retaining raw files alongside exported tables makes discrepancies traceable.

# 5. Conclusion

Both methods produced the expected I-V knee and P-V peak. The highest recorded points were 4.728 W at 13.81 V with 40 Ω and 4.826 W at 14.04 V with 26% duty. The endpoint estimates were about 18.23 V and 0.401 A. The qualitative characterization and load-control aims were met; precise DC MPP, thermal stability and efficiency were not established because acquisition mode, endpoint anomalies and operating conditions were not fully verified.